

BANK STRUCTURAL FAILURE CASE · FDIC CERTIFICATE 24735



Silicon Valley Bank Structural Failure Analysis

An 8-phase N-D-C structural analysis of Silicon Valley Bank using public FDIC Call Report data and SEC filings through Q4 2022. The model identifies the failure 18 months before the March 2023 bank run, from data that was publicly available throughout.

**The core finding for financial managers**

Standard regulatory metrics (CET1 12%, CAMELS 2, no enforcement actions) showed SVB as a healthy, well-capitalized institution through Q4 2022. The N-D-C structural analysis shows Phase 3 (Interest Rate Risk) entered Breakdown in Q3 2021 and remained there for six consecutive quarters. The failure was structurally visible. It was not detected because standard frameworks do not evaluate cross-phase structural balance.

SYSTEM BALANCE (Π)

61.3%

8-phase N-D-C chain

PHASE 3 BALANCE

21.4%

Interest rate risk — Breakdown

BOTTLENECK PHASES

4

Phases 2, 3, 4, 6 below threshold

QUARTERS IN BREAKDOWN

6

Phase 3, Q3 2021 through Q4 2022

THE SIGNAL THE MODEL WOULD HAVE PRODUCED

Phase 3: Interest Rate Risk — Quarterly Balance Score

SVB dissolved its interest rate hedges in Q2 2021 to boost reported net interest income. From that point, Phase 3 (Interest Rate Risk Management) entered Breakdown and never recovered. The bars below show

the balance score each quarter. The failure occurred in Q1 2023, 18 months after the signal.



Q3 2021: SVB removes all interest rate swaps. HTM bond portfolio reaches 55% of assets. Phase 3 crosses the 38.2% Breakdown threshold and does not recover. Standard regulatory capital ratios remain green throughout this entire period. The Fed examination did not escalate concerns with sufficient urgency.

March 10, 2023: Bank run. FDIC takes receivership.

PHASE 0

Deposit Franchise Quality

63.6%

STRESSED

PHASE 1

Capital Adequacy (Reported)

97.5%

DECEPTIVE SIGNAL

PHASE 2

Asset Allocation

29.6%

BREAKDOWN

PHASE 3

Interest Rate Risk

21.4%

BREAKDOWN — PRIMARY

PHASE 4

Liquidity Coverage

47.4%

BREAKDOWN

PHASE 5

Earnings and NIM

81.3%

GOOD

PHASE 6

Market Confidence

57.1%

CRITICAL

PHASE 7

Regulatory Compliance

92.5%

DECEPTIVE SIGNAL

WHAT EACH FRAMEWORK SAID IN Q4 2022

FRAMEWORK	SIGNAL	WHAT IT MISSED
Regulatory capital (CET1)	PASS — 12.1%, well above 10% threshold	AOCI exclusion for mid-tier banks hid \$17.7B unrealized HTM losses; true equity was

FRAMEWORK	SIGNAL	WHAT IT MISSED
		negative
FDIC CAMELS examination	COMPOSITE 2 — satisfactory	Supervisory concerns identified but not escalated; deference to management recovery plans
Credit agency ratings	REVIEW — Moody's placed on review Dec 2022	Review initiated 90 days before failure; too late for most counterparties to act
LCR / liquidity coverage	EXEMPT — mid-tier bank regulatory carve-out	2019 tailoring rules eliminated binding liquidity constraint for banks \$50-250B; no LCR requirement
N-D-C structural analysis	BREAKDOWN — Phase 3 at 21.4% for 6 quarters	Nothing hidden. Phase 3 Breakdown detected from public data in Q3 2021 and sustained through failure

FIVE-MODEL COHERENCE PANEL — SVB Q4 2022



Scores derived from public FDIC Call Report data (cert 24735), SVB 10-K filings, and Federal Reserve SVB supervision review (April 2023). Chain average (π) is the arithmetic mean of 8 phase balances. ϕ , $\sqrt{2}$, $\ln 2$, e scores are structural assessments benchmarked against the five fundamental constants.



Structural Analysis

8-phase N-D-C breakdown with quarterly trend data for Phase 3

- Phase balance scores from public FDIC data
- Quarterly Phase 3 decline from Q1 2021
- D-C imbalance diagnosis per phase
- Comparison to regulatory framework signals

[View Analysis](#)

[Dashboard](#) [Intervention Analysis](#)



Resolution Analysis

FDIC receivership, capital recovery rates, and what could have been recovered with earlier intervention

- Deposit recovery rates by class
- Asset recovery under FDIC resolution
- FDIC insurance fund impact
- Counterfactual: early intervention outcome

[View Resolution](#)

Financial Context

Regulatory reform implications, peer bank screening framework, and forward-looking application

- How to screen peer banks using public data
- Post-SVB regulatory reform scorecard
- Banks with similar structural profiles
- Framework for portfolio-level monitoring

[View Context](#)

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